

From Micro to Macro: Learning Real-Time Economic Signals from Firm-Level Accounting Data^{*}

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May 13, 2026

Abstract

How can real-time information about aggregate economic activity be extracted from firm-level accounting data? This paper proposes a *micro-to-macro* nowcasting approach that uses machine learning algorithms to directly exploit the full cross-section of accounting information from 21,061 publicly listed U.S. firms for nowcasting U.S. aggregate output. Rather than relying on aggregated corporate indicators, the proposed framework preserves the rich information contained in firm-level heterogeneity and cross-firm interactions that is often lost through aggregation. The empirical results show that the proposed approach significantly improves nowcasting accuracy, reducing the root mean squared error by more than 70% relative to a random walk benchmark and substantially outperforming models based on aggregated accounting data. Firm-level corporate accounting data also improve nowcasting performance by about 17.35% relative to models using mixed-frequency aggregate macroeconomic and financial predictors. Our feature-importance analysis shows that the firms with larger assets, lower book-to-market ratios, or lower leverage, and the firms in

^{*}We are grateful for comments and suggestions from Andrii Babii, Raffaella Giacomini, Jia Li, and participants at 2025 International Symposium on Forecasting (Beijing, China), 2025 National Conference on the Digital Economy (Beijing, China), 2025 China Econometricians Forum (Changsha, China), 2025 XMU-CEA (UK/Europe) Symposium on AI and the Digital Economy (Xiamen, China), 2026 RIBAF Symposium on Economic Statistics (Nanchang, China), and 2026 Spring Conference on China's Macroeconomic Forecasting and Analysis (Kuala Lumpur, Malaysia). Hong is supported by MOE Social Sciences Innovative Group on Complex Systems Modeling in Economic Management in the Era of Digital Intelligence, University of Chinese Academy of Sciences (Grant No. E5820801) and National Science Foundation of China (NSFC) (Grant No. 71988101). Huang is supported by the National Science Foundation of China (NSFC) (Grant No. 72473166).

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the industrials, utilities, and consumer discretionary sectors have greater importance for nowcasting U.S. GDP. Furthermore, although the number and composition of the firms that release quarterly financial reports differ over time before the end of a quarter, the GDP nowcasts remain quite stable. Overall, the results underscore the value of directly exploiting firm-level accounting data from all publicly listed firms for nowcasting aggregate economic activity.

Keywords: GDP Nowcasting, Firm-Level Accounting Data, Firm Heterogeneity, Aggregation, Machine Learning

JEL code: C53, C55, E01, E60, M41

1 Introduction

Timely measurement of macroeconomic conditions is crucial for economic decision-making by both policymakers and economic agents (e.g., [Aruoba and Diebold 2010](#), [Aruoba, Diebold, and Scotti 2009](#)). In practice, however, key macroeconomic statistics are released with substantial delays, particularly for low-frequency indicators such as gross domestic product (GDP). These informational frictions highlight the distinction between forecasting and nowcasting. While forecasting targets future economic outcomes, nowcasting focuses on assessing current conditions in real time, providing estimates of the present, the very near future, or the recent past ([Bańbura, Giannone, Modugno, and Reichlin 2013](#)). By exploiting timely, often higher-frequency information and accounting for the flow of data releases, macroeconomic nowcasting helps bridge this information gap by producing early estimates of economic activity before official data become available (e.g., [Andreou, Ghysels, and Kourtellis 2013](#), [Antolín-Díaz, Drechsel, and Petrella 2024](#), [Babii, Ghysels, and Striaukas 2022](#), [Barnett, Chauvet, and Leiva-Leon 2016](#), [Clements and Galvão 2008, 2009](#), [Giannone, Reichlin, and Small 2008](#), [Wright 2019](#)).

In recent years, accounting information of listed firms has been increasingly used in macroeconomic forecasting and nowcasting because of its timeliness and close connection to aggregate economic activity. Unlike traditional macroeconomic variables, which are often subject to publication delays and revisions, corporate financial reports—released quarterly and containing measures such as earnings, revenues, and returns on capital—provide timely signals that reflect real-time conditions at the corporate level. These signals are closely linked to fluctuations in macroeconomic output. In a seminal study, [Konchitchki and Pata-toukas \(2014a\)](#) show that aggregate accounting earnings contain predictive information for

future GDP growth, establishing a direct link between accounting information and macroeconomic activity, and [Konchitchki and Patatoukas \(2014b\)](#) provide evidence that accounting profitability offers predictive content for GDP beyond that reflected in stock returns. [Nallareddy and Ogneva \(2017\)](#) find that earnings dispersion adds incremental value in predicting future GDP growth. Studies such as [Kalay, Nallareddy, and Sadka \(2018\)](#), [Kim, Pandit, and Wasley \(2016\)](#), and [Konchitchki, Luo, Ma, and Wu \(2016\)](#) highlight the correlation between accounting information and future macroeconomic shocks. [Ball, Gallo, and Ghysels \(2019\)](#) show that smoother earnings improve GDP prediction, while [Gaertner, Kausar, and Steele \(2020\)](#) report that negative earnings changes are more informative for future GDP than positive changes. [Abdalla and Carabias \(2022\)](#) show that aggregate special items provide timely signals for real GDP growth, and [Abdalla, Carabias, and Patatoukas \(2021\)](#) show that sector-level accounting data enhance nowcasting accuracy within the canonical dynamic factor nowcasting framework of [Giannone, Reichlin, and Small \(2008\)](#). More recently, [Gallo, Jin, and Sridharan \(2026\)](#) further examine the relationship between aggregate earnings and GDP and investigate how aggregate accounting measures signal future economic conditions. Collectively, these studies underscore the value of accounting data as a complementary source of information, offering timely predictive signals that are not fully captured by conventional macroeconomic or aggregate predictors.

In the existing literature, firm-level accounting information is typically aggregated into higher-level measures before being used for macroeconomic forecasting or nowcasting. For example, [Konchitchki and Patatoukas \(2014a,b\)](#) aggregate firm-level accounting variables into value-weighted market-level averages. Similarly, [Abdalla, Carabias, and Patatoukas \(2021\)](#) construct sector-level aggregates and incorporate them into a dynamic factor model (*DFM*) for GDP nowcasting.

However, aggregation of firm-level accounting data inevitably leads to information loss ([Feige and Watts 1972](#), [Orcutt, Watts, and Edwards 1968](#)). Mechanically, these aggregation approaches compress the cross-sectional distribution of firm outcomes into a low-dimensional representation, eliminating firm heterogeneity, cross-firm interactions, nonlinear dynamics, and sudden structural changes, thereby reducing the effective information available for estimation and prediction. This raises a key question at the heart of our study: how can new real-time economic signals be learned from firm-level accounting data for macroeconomic nowcasting, while preserving the richness of firm-level information and minimizing the information loss inherent in conventional aggregation approaches?

To address this question, we propose a *micro-to-macro* nowcasting approach that learns real-time economic signals from a large panel of firm-level accounting data covering all 21,061

publicly listed U.S. firms to nowcast the U.S. aggregate output. Unlike conventional approaches that aggregate firm-level variables into sector- or market-level measures or into latent factors via factor extraction, the proposed approach operates directly on the large firm-level panel, retaining the richness of micro-level accounting information. We employ machine learning methods to exploit firm-level heterogeneity and interactions. Through regularization and related techniques, these methods can effectively handle high-dimensional panels and capture both linear and nonlinear relationships, thereby minimizing the information loss inherent in conventional aggregation approaches.

More specifically, we construct a predictive panel based on three firm-level accounting variables that are widely used in the literature to forecast or nowcast macroeconomic activity: return on net operating assets (*RNOA*), revenue growth (*Growth*), and accounting earnings (*Earnings*). These measures provide informative signals of corporate activity while capturing complementary dimensions of firm performance—profitability, growth, and operating outcomes. *RNOA* captures operating profitability and capital efficiency, measuring how effectively firms generate income from invested capital and thus serving as a proxy for supply-side productivity. *Growth* reflects expansion in sales and demand conditions, providing information about changes in output and market activity on the demand side. *Earnings*, in turn, summarizes overall profit performance, thereby capturing broader corporate profit conditions. Using these firm-level accounting measures for all publicly listed U.S. firms yields a large predictive panel. We then apply a range of machine learning methods to this large panel to nowcast U.S. real GDP. Our main findings are as follows:

First, our *micro-to-macro* nowcasting approach substantially improves the accuracy of real GDP growth nowcasts relative to the random walk (*RW*) benchmark, reducing RMSE by more than 70% on average. Among alternative variable combinations, the pair of *RNOA* and *Earnings* performs best, achieving the lowest average RMSE ratio of 0.287, while adding a third accounting variable yields little further improvement. This suggests that, in accounting-based nowcasting, more accounting variables are not necessarily better; a parsimonious choice may be sufficient to capture the most informative dimensions of firm fundamentals. In addition, our method works well even early in the quarter, when more than 40% of firms have yet to report accounting information for the previous quarter, and nowcasts are already nearly as accurate as those at the end of the quarter. This highlights an important advantage of the *micro-to-macro* approach, namely its ability to extract useful real-time macroeconomic signals from limited firm-level information available early in the quarter. These gains remain robust during periods of heightened economic uncertainty and volatility, including the 2008 global financial crisis and the COVID-19 pandemic.

Second, to quantify how much information is lost when firm-level accounting signals are aggregated, we compare our *micro-to-macro* nowcasts with those based on aggregated accounting data, including value-weighted market- or sector-level measures and dynamic factor model (*DFM*)-extracted factors. Our approach delivers substantial gains over these alternatives across all variable combinations, highlighting the value of exploiting firm-level information directly rather than through aggregation. Specifically, the *micro-to-macro* nowcasts reduce RMSE by 25.13% over market-level aggregates, 15.21% over sector-level aggregates, and 46.52% over the *DFM*-based factors. Moreover, among value-weighted accounting aggregates, nowcasting performance improves as the level of aggregation decreases: sector-level aggregates outperform market-level aggregates, while firm-level data perform best overall. These results underscore a key insight of our paper: exploiting firm-level accounting data at the micro level is crucial for improving nowcasting performance, as it preserves heterogeneity across firms and interactions among them that are largely masked by aggregation.

Third, we assess whether firm-level accounting variables contain incremental information for nowcasting over and above that captured by mixed-frequency macroeconomic and financial predictors. We use a macro-financial dataset that is widely used in the macroeconomic nowcasting literature and provides broad coverage of U.S. economic conditions, spanning real activity, consumption, labor markets, prices, housing and construction, external trade, and financial conditions. Our results show that firm-level accounting data deliver superior nowcasting performance, with a 15.99% reduction in RMSE over macroeconomic and financial predictors. While combining the two information sets yields a 16.18% reduction, it provides little improvement over accounting data alone, suggesting limited incremental value of aggregate predictors once firm-level information is incorporated.

Lastly, we examine feature importance of individual firms in economic output nowcasting. Firm-level importance is highly heterogeneous and varies over time, with limited overlap early in the quarter and greater stability toward the end of the quarter. When firms are sorted separately by characteristics, larger firms, firms with lower book-to-market ratios, and firms with lower leverage account for a large share of total importance. At the sectoral level, *industrials*, *utilities*, and *consumer discretionary* play especially important roles. Over time, *industrials* dominate early in the sample, while *utilities* rise steadily after the global financial crisis and become the leading sector by around 2013. These findings further highlight the value of using firm-level accounting data, as the importance of individual firms varies across firm characteristics, sectors, and over time in ways that are masked by aggregation, making it harder to identify the firms most relevant for nowcasting.

The remainder of the article is organized as follows. Section 2 describes our *micro-to-*

macro nowcasting approach. Section 3 details the data construction and sample. Section 4 reports the main empirical results on GDP nowcasting performance. Section 5 compares our nowcasts with those based on aggregate accounting variables. Section 6 evaluates individual accounting predictors alongside mixed-frequency macroeconomic and financial predictors. Section 7 examines the cross-sectional importance of firms in nowcasting. Section 8 concludes. Additional results and supporting materials are provided in the Appendix.

4 Main Results

In this section, we present the empirical results to evaluate the effectiveness of our *micro-to-macro* nowcasting approach. Specifically, we assess the extent to which incorporating individual firm-level accounting data improves GDP nowcasting performance relative to the benchmark model. We examine both overall (static) predictive accuracy and time-varying performance, with particular attention to periods of macroeconomic instability such as recessions, the global financial crisis, and the COVID-19 pandemic.

4.1 Comparison with the Benchmark Model

Table 1 reports out-of-sample root mean squared error (RMSE) ratios for our *micro-to-macro* nowcasting models relative to the *random walk* (*RW*) benchmark for U.S. real GDP growth. The use of the *RW* benchmark is consistent with prior literature (e.g., Abdalla et al. 2021, Andreou et al. 2010, Giannone et al. 2008).¹ An RMSE ratio below one indicates that a model outperforms the *RW* benchmark. Each panel reports results for the best-performing model—defined as the model with the lowest average RMSE across algorithms—within each variable specification. In most cases, *Random Forest* (*RF*) delivers the best overall performance among all machine learning algorithms considered.² Across all specifications, the average RMSE ratios range from 0.29 to 0.31, implying roughly a 70% reduction in forecasting error relative to the *RW* model. These findings underscore the value of our *micro-to-macro* approach, which fully uses the rich cross-sectional information embedded in the firm-level ac-

¹Formal definitions of RMSE and the *RW* model are provided in Appendix A.2.

²*RF* is a nonlinear and relatively dense machine learning algorithm that is well suited to capturing interaction effects and complex patterns across individual firms, which contributes to its strong nowcasting performance. This result is in line with Cui et al. (2026), Giannone et al. (2021), and Shen and Xiu (2025), who likewise emphasize the advantages of dense and non-linear algorithms in high-dimensional macroeconomic forecasting settings. Detailed comparisons across all machine learning algorithms are provided in Appendix C.1.

Table 1: RMSE Ratios Relative to the *RW* Model: U.S. Real GDP Growth

Variable Specification	Information Cutoff Date									
	$t=-8$	$t=-7$	$t=-6$	$t=-5$	$t=-4$	$t=-3$	$t=-2$	$t=-1$	$t=0$	Avg.
Panel (a): Single Accounting Variable										
<i>RNOA</i>	0.296	0.302	0.297	0.292	0.296	0.296	0.292	0.290	0.298	0.296
<i>Growth</i>	0.307	0.318	0.303	0.299	0.301	0.299	0.303	0.304	0.296	0.303
<i>Earnings</i>	0.298	0.299	0.307	0.311	0.312	0.320	0.313	0.325	0.311	0.311
Panel (b): Two Accounting Variables										
<i>RNOA–Growth</i>	0.291	0.291	0.291	0.291	0.294	0.294	0.297	0.294	0.294	0.293
<i>RNOA–Earnings</i>	0.287	0.285	0.281	0.285	0.289	0.289	0.289	0.289	0.289	0.287
<i>Growth–Earnings</i>	0.302	0.299	0.303	0.303	0.297	0.298	0.299	0.297	0.298	0.300
Panel (c): Three Accounting Variables										
<i>RNOA–Growth–Earnings</i>	0.287	0.289	0.287	0.292	0.291	0.295	0.293	0.292	0.294	0.291

Notes: The table reports, for each information cutoff date t , RMSE ratios for nowcasts of U.S. real GDP growth generated by each *micro-to-macro* nowcasting model relative to the *RW* benchmark model over the out-of-sample period (2000:Q3–2024:Q2). “Avg.” denotes the average RMSE ratio across all information cutoff dates. Each panel reports results for the best-performing model (i.e., the model with the lowest average RMSE across machine learning algorithms) within the corresponding variable specification. Panel (a) reports results for three single-variable models based on *RNOA*, *Growth*, and *Earnings*. Panel (b) reports results for three two-variable combinations: *RNOA–Growth*, *RNOA–Earnings*, and *Growth–Earnings*. Panel (c) reports results for the three-variable model combining *RNOA*, *Growth*, and *Earnings*. The RMSE ratio of the best-performing model is highlighted in bold. Corresponding RMSE values are reported in Appendix C.1.

counting panel by exploiting heterogeneity and interaction information across firms, thereby substantially improving GDP nowcasting accuracy over the *RW* benchmark.

Examining performance across different information cutoff dates, we find no clear evidence that RMSE ratios systematically decline as the quarter progresses. This highlights a key advantage of the *micro-to-macro* approach: even at $t = -8$, when more than 40% of firms have yet to report accounting information for the previous quarter, nowcasts based on the available firm-level information perform nearly as well as those constructed at $t = 0$. This suggests that much of the useful predictive content is already available early in the quarter. A plausible explanation is that firms are exposed to common macroeconomic and industry shocks and are connected through production and information networks. As a result, accounting information released by early-reporting firms can also be informative about firms that report later. Moreover, early reporters are not necessarily a random subset of firms. They are often bellwether firms within their industries, and their disclosures may therefore contain disproportionately important information about sectoral and aggregate economic conditions (Foster 1981). This interpretation is consistent with the granular view of aggregate fluctuations: in economies with fat-tailed firm-size distributions, the performance of

large firms can have non-negligible effects on aggregate activity (Carvalho and Gabaix 2013, Gabaix 2011). It is also consistent with the network view that firm- and sector-level shocks can propagate through input-output linkages, further amplifying the macroeconomic relevance of information released by central or bellwether firms (Acemoglu et al. 2012). In this sense, early disclosures may partly substitute for information that becomes available only later, enabling the model to produce accurate nowcasts well before the end of the quarter, even when reporting coverage remains incomplete.

Comparisons across specifications reveal several important patterns. Among the single-variable models, *RNOA* delivers the lowest average RMSE ratio, followed closely by *Growth*, with *Earnings* performing slightly worse. This suggests that operating profitability contains somewhat stronger contemporaneous signals than revenue or earnings growth when considered alone. The differences, however, are statistically small.

Moving from one-variable to two-variable specifications generally improves performance. The combination of *RNOA* and *Earnings* performs best overall, achieving the lowest average RMSE ratio (0.287), corresponding to about a 71% reduction over the *RW* benchmark.³

In contrast, adding the third accounting variable does not yield substantial additional gains. The three-variable specification achieves an average RMSE ratio of 0.291, which is slightly worse than the best two-variable model. This finding indicates that most of the relevant macroeconomic information embedded in firm-level accounting data can already be captured by a parsimonious combination of *RNOA* and *Earnings*, with limited incremental benefit from further including *Growth*.

The strong nowcasting performance of *RNOA*-based models is economically intuitive. *RNOA* captures operating profitability relative to invested capital and thus reflects firms' production efficiency. At the aggregate level, GDP growth reflects value added from the utilization of labor and capital, making *RNOA* a micro-level proxy for aggregate productivity. Moreover, firm profits contribute to aggregate corporate profits, which constitute an important component of economic output and income (Fischer and Merton 1984, Marcuss 2004), implying that firm-level operating profitability provides timely signals of aggregate economic performance. The predictive value of *RNOA* for aggregate output is also supported by prior evidence (Abdalla et al. 2021, Konchitchki and Patatoukas 2014b).

The large predictive gains from combining *RNOA* and *Earnings* likely reflect the com-

³The superior performance of the *RNOA*–*Earnings* combination is consistent with the Model Confidence Set (MCS) results reported in Table C.4 in Appendix C.1, where the specification is selected as the best-performing model for eight out of nine information cutoff dates.

plementary information captured by these two variables. While *RNOA* summarizes firms’ operating efficiency and production-side performance, *Earnings* captures broader bottom-line profitability that incorporates revenue changes, cost adjustments, financial conditions, and other non-operating components. Prior evidence suggests that the predictive power of earnings for aggregate economic activity operates through several macroeconomic channels, including aggregate corporate profits (e.g., [Abdalla and Carabias 2022](#), [Konchitchki and Patatoukas 2014a](#)), inflation ([Shivakumar and Urcan 2017](#)), monetary policy ([Gallo et al. 2016](#)), and the labor market (e.g., [Gallo et al. 2026](#), [Hann et al. 2021](#), [Nallareddy and Ogneva 2017](#)). Taken together, these variables capture both operational efficiency and broader income dynamics that are closely linked to aggregate economic output.

The limited incremental gain from adding *Growth* to the *RNOA–Earnings* specification likely reflects both the nature of the information captured by revenue growth and its overlap with the other two variables. *Growth* measures top-line revenue growth and therefore mainly captures changes in demand conditions and other short-run fluctuations faced by firms, consistent with prior studies that interpret firm-level sales dynamics in this way (e.g., [Bloom et al. 2007](#), [Guiso and Parigi 1999](#), [Love and Zicchino 2006](#)). This helps explain why the *RNOA–Growth* combination performs reasonably well. However, compared with *RNOA*, revenue growth is a less direct measure of economic value creation because it does not account for operating costs or the capital employed to generate sales. Moreover, the marginal contribution of *Growth* is limited because much of the economic information it contains is already captured by *RNOA* and *Earnings*. Sales growth affects operating income and is therefore reflected in *RNOA* through operating profitability, while its effects on net income—after operating costs, financing expenses, taxes, and other adjustments—are captured by *Earnings*. As a result, once *RNOA* and *Earnings* are jointly included, *Growth* adds little incremental information for nowcasting GDP.

4.2 Time-Varying Predictability

Accurately nowcasting GDP growth is particularly challenging during periods of macroeconomic instability—such as recessions, the 2008 global financial crisis, and the COVID-19 pandemic—when the economic environment becomes more complex and less predictable. We now conduct an analysis to investigate the effectiveness of our *micro-to-macro* nowcasting approach during high-volatility periods. The recession periods, as identified by the NBER, are defined as 2008:Q1–2009:Q2 and 2020Q1–2020:Q2. In addition, we analyze two broader periods of instability: the global financial crisis spanning from 2007:Q2 to 2010:Q2, and the

COVID-19 period ranging from 2020:Q1 to 2022:Q1.

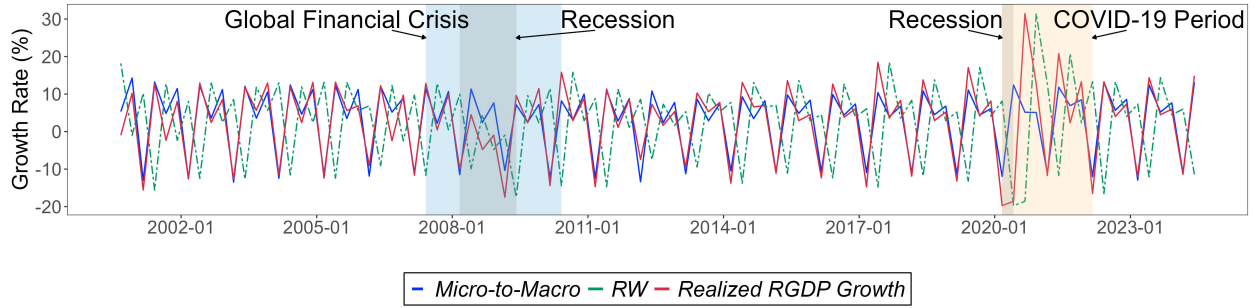


Figure 1: Realized U.S. Real GDP Growth and Nowcasts from *RW* and *Micro-to-Macro* Nowcasting Models: *RNOA-Earnings*

Notes: The figure reports realized U.S. real GDP growth, nowcasts from the *RW* benchmark, and nowcasts at $t = 0$ from the *micro-to-macro* nowcasting approach. The *micro-to-macro* nowcasts are based on the *RF* model, which has the lowest average RMSE for real GDP growth among the machine learning algorithms, and on the *RNOA-Earnings* specification, which performs best among the accounting variable combinations. NBER recession periods, the global financial crisis, and the COVID-19 period are highlighted by gray, blue, and orange shading, respectively.

Figure 1 plots realized U.S. real GDP growth together with nowcasts from the *RW* benchmark and the *micro-to-macro* approach, allowing a direct comparison of their time-varying nowcasting performance. Visually, the *micro-to-macro* approach tracks realized GDP growth more closely over time, with forecast deviations that are generally smaller than those of the *RW* benchmark. This advantage is especially pronounced during periods of high volatility, including recessionary episodes and the COVID-19 period, when the *RW* benchmark exhibits substantially larger forecast misses. The figure also suggests that the *micro-to-macro* approach performs particularly well in episodes of negative growth, where accurately assessing the extent of economic deterioration is especially important. In particular, for observations with negative realized GDP growth during high-volatility periods, the average forecast error, defined as forecast minus actual growth, is 7.39% for the *micro-to-macro* approach, compared with 16.53% for the *RW* benchmark. Thus, although both models tend to overpredict growth during downturns, the degree of overprediction is much smaller for our method.

We also compute the average absolute forecast errors of the two models. The *micro-to-macro* approach consistently outperforms the *RW* benchmark during periods of high volatility. Specifically, the average absolute forecast error of the *micro-to-macro* approach is approximately 65.21% lower than that of the *RW* benchmark in these periods.⁴ Taken together,

⁴The results reported for absolute forecast errors in high-volatility periods and forecast errors in negative-growth episodes are averaged over all information cutoff dates. Detailed results for each information cutoff date are reported in Appendix C.2.

the evidence from both the absolute forecast errors and forecast errors in negative-growth episodes points to the same conclusion: incorporating firm-level accounting information substantially improves real-time macroeconomic nowcasting, particularly when macroeconomic conditions are volatile and downside risks are most relevant.

4.3 Robustness Check

We examine the robustness of our results by nowcasting nominal GDP growth. The *micro-to-macro* models deliver substantial improvements over the *RW* benchmark, with RMSE reductions of about 70%. We next consider real gross domestic income (GDI) growth, an alternative measure of economic activity representing the income side of the BEA’s NIPA. Prior research, such as [Nalewaik \(2010\)](#), highlights the value of GDI as a complementary indicator to GDP. Our approach again delivers clear improvements in nowcasting accuracy over the benchmark. Due to space limitations, all these results are reported in [Appendix C.3](#). Overall, the results are consistent with those for real GDP growth, supporting the reliability and generalizability of the *micro-to-macro* approach.

7 Feature Importance

In this section, we examine the feature importance of firm-level accounting data for GDP nowcasting. Specifically, we assess the heterogeneous contributions of individual firms to GDP nowcasts, investigate how importance varies across firm characteristics, and explore differences across industry sectors. This analysis underscores a key advantage of our *micro-to-macro* approach over existing methods, especially those based on aggregation. Because our approach retains firm-level information, it enables us to identify the firms most relevant for GDP nowcasting and to examine the firm characteristics associated with their importance. This, in turn, offers a new empirical insight into how heterogeneous firms contribute to aggregate macroeconomic outcomes.

The analysis focuses on models using *RNOA* and *Earnings*, which deliver the best overall nowcasts for real GDP growth. All importance results are based on the *RF* model, which emerges as the top-performing algorithm under this accounting-variable specification.⁵

⁵Details on the calculation of feature importance are provided in [Appendix B.2](#).

7.1 Firm-Level Analysis

- **Heterogeneous Importance Across Firms**



Figure 2: Word Cloud of Important Firms at $t = 0$

Notes: The figure displays the tickers of the 200 firms with the highest feature importance over the out-of-sample period for $t = 0$.

We first examine the top 200 firms with the largest contributions at each information cutoff date.⁶ Figure 2 visualizes firm importance at the information cutoff date $t = 0$, when the accounting information set is most complete. Each word in the cloud corresponds to a listed firm’s ticker, with its size reflecting the relative importance of that firm’s accounting information for GDP nowcasts. For example, the relatively large size of “OMC” indicates that the accounting information of *Omnicom Group Inc.* contributes substantially to GDP nowcasts.

For a given information cutoff date, the contributions of individual firms differ substantially. For example, at $t = 0$, the wide variation in word sizes in the word cloud indicates pronounced heterogeneity in firm importance. Specifically, the most influential firm accounts for approximately 3.06% of total importance, while the next four firms each contribute between 1.10% and 0.73%. In contrast, the firm ranked 200th contributes only about 0.13%, implying an importance level less than one-twentieth of that of the top-ranked firm.

The composition of important firms also varies substantially as new accounting information becomes available. Table 2 reports the pairwise overlap of firms in the top-200 importance rankings across different information cutoff dates. The set of top 200 firms at $t = 0$ differs markedly from that at $t = -8$, with only 99 firms appearing in both rankings. For example, at the early cutoff date $t = -8$, the top five firms include *Omnicom Group Inc.*, *Universal Health Services Inc.*, *Dover Corp.*, *Penske Automotive Group Inc.*, and *McDonald's Corp.* In contrast, at the quarter-end cutoff $t = 0$, the most influential

⁶Due to space limitations, the full list of the top 200 firms and their corresponding feature importance percentages are reported in Table C.14 in Appendix C.6.

firms shift to *Omnicom Group Inc.*, *Alabama Power Co.*, *Six Flags Entertainment Corp.*, *Sherwin-Williams Co.*, and *Lafarge North America Inc.*, reflecting a substantially different firm composition. At the same time, the overlap increases as more accounting information becomes available: at $t = -4$, as many as 181 firms in the top-200 list also appear at $t = 0$. This suggests that the composition of influential firms becomes more stable as corporate financial statements are released and accounting information accumulates over the quarter.

Table 2: Pairwise Overlap of Top 200 Most Important Firms by Information Cutoff Date

Information Cutoff Date	Information Cutoff Date		
	$t = -8$	$t = -4$	$t = 0$
$t = -8$	200	—	—
$t = -4$	100	200	—
$t = 0$	99	181	200

Notes: The table reports the pairwise overlap of firms in the top-200 lists at $t \in \{-8, -4, 0\}$. The rows and columns represent information cutoff dates. Each entry shows the number of firms in the top-200 list for the row cutoff date which also appear in the top-200 list for the column cutoff date.

Moreover, across all information cutoff dates, we find that 362 distinct firms appear at least once in the top-200 importance rankings, while only 83 firms—22.93% of the total—are selected at all cutoff dates. This result further indicates that firms’ contributions to GDP nowcasts change over time. The set of key contributing firms evolves as new information arrives, rather than remaining the same throughout the quarter. This highlights the flexibility of our *micro-to-macro* approach in accommodating time-varying predictability. As the information environment evolves, our approach allows the relative importance of individual firms to adjust dynamically with the flow of accounting disclosures.

• Importance by Firm Characteristics

Next, we assess how individual firms’ nowcasting importance varies with firm characteristics. Firms are separately sorted, for each firm characteristic, into two equally sized groups—*Top 1/2* and *Bottom 1/2*—based on cross-sectional rankings. These characteristics include *firm size*, *book-to-market (BM) ratio*, and *leverage*. *Firm size* captures differences in the scale of firms’ economic activity, *BM ratio* reflects cross-sectional variation in firms’ valuation relative to their accounting fundamentals, and *leverage* characterizes differences in capital structure and reliance on debt financing.⁷

⁷*Firm size* is measured as the natural logarithm of market capitalization, which requires market capitalization to be positive. As a robustness check (not reported in the paper), we alternatively use the natural

Table 3: Group-Level Feature Importance by Firm Characteristics

Firm Characteristic	Category	Information Cutoff Date									Avg.
		$t = -8$	$t = -7$	$t = -6$	$t = -5$	$t = -4$	$t = -3$	$t = -2$	$t = -1$	$t = 0$	
Firm Size	Top 1/2	67.78%	61.63%	58.85%	58.59%	57.59%	60.46%	61.01%	61.06%	60.35%	60.81%
	Bottom 1/2	17.45%	19.63%	20.52%	21.09%	23.05%	19.62%	18.50%	17.92%	18.09%	19.54%
BM Ratio	Top 1/2	23.68%	24.58%	23.29%	23.77%	25.73%	26.18%	26.09%	25.64%	25.25%	24.91%
	Bottom 1/2	59.78%	54.86%	54.37%	54.06%	52.98%	51.80%	51.22%	51.14%	51.00%	53.47%
Leverage	Top 1/2	40.72%	38.22%	36.72%	36.29%	34.49%	36.62%	36.47%	36.35%	36.17%	36.89%
	Bottom 1/2	46.12%	47.33%	48.03%	46.72%	46.95%	43.75%	43.47%	43.24%	43.22%	45.43%

Notes: This table reports group-level importance, expressed as percentages of total importance, for firms sorted by selected characteristics. The results are based on the *RF* model, which is the best-performing *micro-to-macro* nowcasting model under the *RNOA-Earnings* specification. For each characteristic, firms are split into two equally sized groups based on cross-sectional rankings: *Top 1/2* and *Bottom 1/2*. The *AR* terms and the *Other* category are excluded; therefore, the reported shares do not sum to 100%.

Table 3 reports the importance percentages of characteristic-sorted firm groups at each information cutoff date. When firms are grouped by *firm size*, larger firms consistently contribute more to GDP nowcasting. Firms in the top half of the size distribution account for approximately 60% of total importance at all information cutoff dates. This difference is particularly pronounced at earlier cutoff dates, where larger firms account for 67.78% of total importance at $t = -8$. This pattern may reflect the fact that accounting information from larger firms captures broader aggregate economic activity and is released more promptly. As a result, such firms play a more important role, especially early in the quarter when accounting information is relatively sparse.

We also find clear differences when firms are grouped by *BM ratio*. Firms with low BM ratios exhibit greater importance over firms with high ratios in GDP nowcasting. On average, the low BM group accounts for 53.47% of total importance, compared with only 24.91% for the high BM group, a gap that remains stable across information cutoff dates. One possible interpretation of this pattern relates to differences in firms' economic conditions and growth prospects. Firms with high BM ratios are often associated with relatively distressed states, with market valuations already incorporating unfavorable expectations about future profitability (Fama and French 1992). In such cases, accounting outcomes are more likely to reflect ex post realizations of adverse macroeconomic conditions, which may limit their usefulness for macroeconomic nowcasting. In contrast, firms with low BM ratios tend to exhibit stronger growth prospects and more persistent expansion, making their operating perfor-

logarithm of total assets and obtain similar results. *BM ratio* is defined as shareholders' equity divided by market capitalization, with both components required to be positive. *Leverage* is measured as the ratio of total interest-bearing debt to total assets, where total assets must be positive.

mance more responsive to short-run fluctuations in aggregate demand (Fama and French 1995). Moreover, because the valuation of low BM firms depends more heavily on future investment opportunities, their performance may be more sensitive to investment-related macroeconomic shocks (Papanikolaou 2011), thereby making accounting information from these firms more informative for real-time macroeconomic nowcasting.

When firms are grouped by *leverage*, lower-leverage firms account for a larger share of total importance at all information cutoff dates, with the difference being modest at early cutoff dates and more pronounced toward the end of the quarter. Prior work shows that firms with lower leverage (lower default risk) respond more strongly and directly to aggregate shocks, as their investment and operating decisions are less distorted by financial frictions, whereas high-leverage firms’ responses are dampened by financing constraints (Ottonello and Winberry 2020). In addition, high leverage can restrict firms’ investment and growth through debt overhang and cash-flow crowding-out effects (e.g., Lang et al. 1996, Myers 1977), limiting their ability to adjust real activities in response to new information. As a result, information from low-leverage firms, which may respond more flexibly, plays a larger role in explaining aggregate outcomes and accounts for a greater share of total importance.

7.2 Sector-Level Analysis

We now examine how feature importance varies across industry sectors. Firms are classified according to the Global Industry Classification Standard (GICS) industry sectors: (1) *energy*; (2) *materials*; (3) *industrials*; (4) *consumer discretionary*; (5) *consumer staples*; (6) *health care*; (7) *financials*; (8) *information technology*; (9) *communication services*; (10) *utilities*; (11) *real estate*.⁸ Sector-level importance is constructed by aggregating firm-level importance within each industry sector.

Table 4 details the importance of each industry sector across different information cutoff dates, where the reported numbers represent the mean relative sectoral importance (as a percentage of the total) averaged over the entire out-of-sample period.

The three most important sectors for nowcasting U.S. real GDP growth are *industrials*, *utilities*, and *consumer discretionary*. Among them, firms in the *industrials* sector stand out with an average importance of around 20% across all information cutoff dates, while firms

⁸Firm-level GICS classifications are not reported in *Compustat Unrestated Quarterly*. We obtain the GICS classification variable from *Compustat Fundamentals Quarterly*. Firms that cannot be matched across the two datasets are assigned to a group labeled *non-classified*. We also include lagged GDP growth rates as a separate category, labeled *AR terms*.

Table 4: Sectoral Importance for U.S. Real GDP Growth Nowcast: *RNOA-Earnings*

Sector	Information Cutoff Date									
	$t = -8$	$t = -7$	$t = -6$	$t = -5$	$t = -4$	$t = -3$	$t = -2$	$t = -1$	$t = 0$	Avg.
Industrials	21.97%	20.08%	20.32%	20.56%	19.05%	18.46%	18.14%	18.28%	18.45%	19.48%
Utilities	8.21%	14.33%	15.78%	15.29%	15.89%	16.18%	14.74%	14.62%	14.59%	14.40%
Consumer Discretionary	16.35%	13.08%	11.84%	10.98%	11.21%	12.00%	12.56%	12.70%	12.51%	12.58%
Financials	8.07%	8.66%	8.40%	8.17%	8.35%	8.52%	8.75%	8.66%	8.90%	8.50%
Health Care	9.76%	9.73%	8.74%	8.12%	7.16%	7.41%	7.49%	7.54%	7.43%	8.15%
Materials	8.60%	7.44%	6.71%	6.74%	6.55%	6.53%	6.69%	6.86%	6.80%	6.99%
Real Estate	3.26%	4.23%	5.38%	6.41%	8.19%	8.65%	8.84%	8.53%	8.67%	6.90%
Communication Services	6.59%	6.05%	5.27%	6.07%	6.17%	5.93%	6.05%	5.90%	5.96%	6.00%
Information Technology	6.25%	5.43%	5.04%	5.09%	4.35%	4.61%	4.65%	4.65%	4.52%	4.95%
Consumer Staples	3.39%	4.03%	5.12%	4.96%	5.03%	3.20%	3.19%	3.24%	3.23%	3.93%
Energy	2.95%	2.85%	3.11%	3.51%	3.84%	4.06%	4.38%	4.46%	4.41%	3.73%
AR Terms	3.47%	3.20%	3.36%	3.31%	3.44%	3.76%	3.80%	3.86%	3.81%	3.56%
Non-Classified	1.13%	0.89%	0.92%	0.80%	0.77%	0.69%	0.72%	0.70%	0.72%	0.82%

Notes: The table reports detailed results for sectoral importance, expressed as shares of total importance in percentage terms, based on the *RF* model, which is the best-performing *micro-to-macro* nowcasting model under the *RNOA-Earnings* specification. The last column represents the mean relative feature importance across all information cutoff dates. The three most important industries are highlighted in bold.

in both *utilities* and *consumer discretionary* also show important contributions, each with an average importance above 10%. The *industrials* sector—including manufacturing, transportation, construction, and infrastructure—accounts for a substantial share of supply-side activity and closely tracks fluctuations in industrial production. The *utilities* sector, providing electricity, natural gas, and water, supports both industrial operations and household consumption, making shifts in utility usage an early signal of changes in overall economic activity. *Consumer discretionary* industries, typically including durable goods, luxury goods, entertainment, leisure activities, and automobiles, are highly sensitive to macroeconomic conditions. Their activity responds strongly to expansions and contractions in the business cycle (e.g., [Attanasio et al. 2022](#), [Baxter 1996](#), [Bils and Klenow 1998](#), [Gomes et al. 2009](#), [Stock and Watson 2002](#)). In contrast, *consumer staples* exhibit much lower elasticity and show little variation over time. As a result, activity in *consumer discretionary* provides timely signals of changes in personal consumption, the largest component of U.S. GDP.

Figure 3 presents the time-varying feature importance of the three leading sectors in nowcasting U.S. real GDP growth during the out-of-sample period.⁹ Firms in the *industrials* sector maintain a leading position in sectoral importance during the earlier part of the out-

⁹The time-varying feature importance is calculated as the average feature importance for each sector across different information cutoff dates.

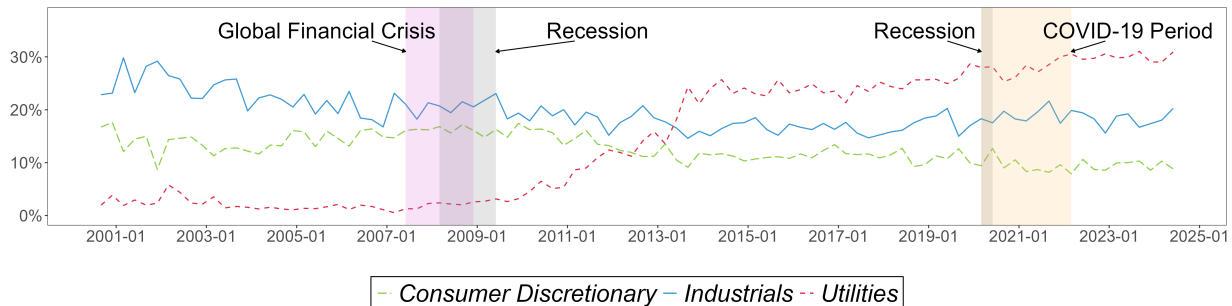


Figure 3: Feature Importance: Top Three Industry Sectors

Notes: The figure depicts the importance of the three leading sectors over the out-of-sample period (2000:Q3–2024:Q2), expressed as percentages of total importance and averaged across information cutoff dates, based on the *RF* model, which is the best-performing *micro-to-macro* nowcasting model under the *RNOA–Earnings* specification. The NBER recession periods, the global financial crisis period, and the COVID-19 period are highlighted in gray, blue, and orange shading, respectively.

of-sample period. The importance of *utilities* increases steadily after the global financial crisis of 2008–2009 and, by around 2013, surpasses *industrials* and *consumer discretionary* to become the most important sector. This leading position is further reinforced during the COVID-19 period.

After the global financial crisis, the U.S. economy entered a prolonged period of exceptionally low interest rates and gradual recovery, a macro-financial environment that was particularly favorable to the *utilities* sector. Given the regulated nature of the industry, relatively inelastic demand, stable cash flows, and large capital expenditure requirements, utility firms were well positioned to expand investment in generation, transmission, and distribution infrastructure under persistently accommodative financial conditions. At the same time, energy infrastructure investment rose markedly in the following decade. Evidence from the U.S. Energy Information Administration (EIA) shows that the growth in utility spending was driven to a large extent by capital investment in electric infrastructure, especially the expansion and upgrading of transmission and distribution systems. These sectoral features may have allowed utility firms’ financial statements to reflect macroeconomic changes in investment, demand, and costs earlier and more steadily than those of more cyclical sectors, thereby increasing the marginal informational value of *utilities* for GDP nowcasting over time.

8 Conclusion

In this paper, we propose a *micro-to-macro* approach that uses a large panel of firm-level accounting data to nowcast aggregate economic output. Our results show that the *micro-to-macro* approach substantially improves nowcasting accuracy for real GDP growth upon the *RW* benchmark, reducing RMSE by more than 70% on average and maintaining strong performance during periods of heightened economic uncertainty. Among alternative accounting variable combinations, the pair of *RNOA* and *Earnings* performs best, while adding a third accounting variable (*Growth*) yields little further improvement. The *micro-to-macro* nowcasting approach performs well early in the quarter, indicating that useful macroeconomic signals can already be extracted from limited firm-level information before reporting coverage becomes complete.

The superior performance over the use of aggregated accounting predictors further highlights the advantage of the *micro-to-macro* approach in preserving firm-level heterogeneity and cross-firm interactions that are typically lost in aggregation. Moreover, compared with mixed-frequency macroeconomic and financial predictors, firm-level accounting data already contain rich and timely signals of aggregate economic activity, while combining the two information sets yields little additional gains.

The feature-importance analysis reveals substantial heterogeneity in firm-level contributions to GDP nowcasting across both firms and information cutoff dates. Firms with larger assets, lower book-to-market ratios, or lower leverage contribute more to GDP nowcasts. At the sector level, firms in *industrials*, *utilities*, and *consumer discretionary* play particularly important roles. These findings highlight an important advantage of the *micro-to-macro* approach: it not only improves GDP nowcasting accuracy but also offers insights into which firms and sectors contain predictive information.

Overall, our study shows that firm-level accounting data contain rich macroeconomic information that can be effectively extracted without prior aggregation. By bridging micro-level corporate behavior and macroeconomic outcomes, our *micro-to-macro* approach offers a novel perspective on real-time economic measurement and a useful tool for assessing current economic conditions.

References

- Abdalla, A. M. and J. M. Carabias (2022). From accounting to economics: The role of aggregate special items in gauging the state of the economy. *Accounting Review* 97(1), 1–27.
- Abdalla, A. M., J. M. Carabias, and P. N. Patatoukas (2021). The real-time macro content of corporate financial reports: A dynamic factor model approach. *Journal of Monetary Economics* 118, 260–280.
- Acemoglu, D., V. M. Carvalho, A. Ozdaglar, and A. Tahbaz-Salehi (2012). The network origins of aggregate fluctuations. *Econometrica : journal of the Econometric Society* 80(5), 1977–2016.
- Andreou, E., E. Ghysels, and A. Kourtellis (2010). Regression models with mixed sampling frequencies. *Journal of Econometrics* 158(2), 246–261.
- Andreou, E., E. Ghysels, and A. Kourtellis (2013). Should macroeconomic forecasters look at daily financial data? *Journal of Business and Economic Statistics* 31(2), 240–251.
- Antolín-Díaz, J., T. Drechsel, and I. Petrella (2024). Advances in nowcasting economic activity: The role of heterogeneous dynamics and fat tails. *Journal of Econometrics* 238(2), 105634.
- Aruoba, S. B. and F. X. Diebold (2010). Real-Time Macroeconomic Monitoring: Real Activity, Inflation, and Interactions. *American Economic Review* 100(2), 20–24.
- Aruoba, S. B., F. X. Diebold, and C. Scotti (2009). Real-Time Measurement of Business Conditions. *Journal of Business and Economic Statistics* 27(4), 417–427.
- Attanasio, O., K. Larkin, M. O. Ravn, and M. Padula (2022). (S)Cars and the Great Recession. *Econometrica* 90(5), 2319–2356.
- Babii, A., E. Ghysels, and J. Striaukas (2022). Machine Learning Time Series Regressions With an Application to Nowcasting. *Journal of Business and Economic Statistics* 40(3), 1094–1106.
- Ball, R. T., L. Gallo, and E. Ghysels (2019). Tilting the evidence: The role of firm-level earnings attributes in the relation between aggregated earnings and gross domestic product. *Review of Accounting Studies* 24(2), 570–592.
- Bañbura, M., D. Giannone, M. Modugno, and L. Reichlin (2013). Now-Casting and the Real-Time Data Flow. In *Handbook of Economic Forecasting*, Volume 2, pp. 195–237. Elsevier B.V.
- Barnett, W. A., M. Chauvet, and D. Leiva-Leon (2016). Real-time nowcasting of nominal GDP with structural breaks. *Journal of Econometrics* 191(2), 312–324.
- Baxter, M. (1996). Are Consumer Durables Important for Business Cycles? *Review of*

- Economics and Statistics* 78(1), 147–155.
- Bils, M. and P. J. Klenow (1998). Using Consumer Theory to Test Competing Business Cycle Models. *Journal of Political Economy* 106(2), 233–261.
- Bloom, N., S. Bond, and J. van Reenen (2007). Uncertainty and Investment Dynamics. *Review of Economic Studies* 74(2), 391–415.
- Carvalho, V. and X. Gabaix (2013). The great diversification and its undoing. *American Economic Review* 103(5), 1697–1727.
- Clements, M. P. and A. B. Galvão (2008). Macroeconomic Forecasting with Mixed-Frequency Data: Forecasting Output Growth in the United States. *Journal of Business and Economic Statistics* 26, 546–554.
- Clements, M. P. and A. B. Galvão (2009). Forecasting US output growth using leading indicators: An appraisal using MIDAS models. *Journal of Applied Econometrics* 24(7), 1187–1206.
- Cui, Y., Y. Hong, N. Huang, and Y. Wang (2026). Forecasting GDP Growth Rates Using Accounting Earnings: A Large Panel Microdata Approach. *Management Science*.
- Fama, E. F. and K. R. French (1992). The Cross-Section of Expected Stock Returns. *Journal of Finance* 47(2), 427–465.
- Fama, E. F. and K. R. French (1995). Size and Book-to-Market Factors in Earnings and Returns. *Journal of Finance* 50(1), 131–155.
- Feige, E. L. and H. W. Watts (1972). An Investigation of the Consequences of Partial Aggregation of Micro-economic Data. *Econometrica*, 343–360.
- Fischer, S. and R. C. Merton (1984). Macroeconomics and Finance: The Role of the Stock Market. Technical Report w1291, National Bureau of Economic Research.
- Foster, G. (1981). Intra-industry information transfers associated with earnings releases. *Journal of Accounting and Economics* 3(3), 201–232.
- Gabaix, X. (2011). The granular origins of aggregate fluctuations. *Econometrica : journal of the Econometric Society* 79(3), 733–772.
- Gaertner, F. B., A. Kausar, and L. B. Steele (2020). Negative accounting earnings and gross domestic product. *Review of Accounting Studies* 25, 1382–1409.
- Gallo, L. A., R. N. Hann, and C. Li (2016). Aggregate earnings surprises, monetary policy, and stock returns. *Journal of Accounting and Economics* 62(1), 103–120.
- Gallo, L. A., H. Jin, and S. A. Sridharan (2026). Unraveling the time-series dynamics between aggregate earnings and GDP. *Journal of Accounting and Economics* 81(1), 101806.
- Giannone, D., M. Lenza, and G. E. Primiceri (2021). Economic Predictions With Big Data: The Illusion of Sparsity. *Econometrica* 89(5), 2409–2437.
- Giannone, D., L. Reichlin, and D. Small (2008). Nowcasting: The real-time informational

- content of macroeconomic data. *Journal of Monetary Economics* 55(4), 665–676.
- Gomes, J. F., L. Kogan, and M. Yogo (2009). Durability of Output and Expected Stock Returns. *Journal of Political Economy* 117(5), 941–986.
- Guiso, L. and G. Parigi (1999). Investment and Demand Uncertainty. *Quarterly Journal of Economics* 114(1), 185–227.
- Hann, R. N., C. Li, and M. Ogneva (2021). Another Look at the Macroeconomic Information Content of Aggregate Earnings: Evidence from the Labor Market. *Accounting Review* 96(2), 365–390.
- Kalay, A., S. Nallareddy, and G. Sadka (2018). Uncertainty and sectoral shifts: The interaction between firm-level and aggregate-level shocks, and macroeconomic activity. *Management Science* 64(1), 198–214.
- Kim, K., S. Pandit, and C. E. Wasley (2016). Macroeconomic uncertainty and management earnings forecasts. *Accounting Horizons* 30(1), 157–172.
- Konchitchki, Y., Y. Luo, M. L. Ma, and F. Wu (2016). Accounting-based downside risk, cost of capital, and the macroeconomy. *Review of Accounting Studies* 21, 1–36.
- Konchitchki, Y. and P. N. Patatoukas (2014a). Accounting earnings and gross domestic product. *Journal of Accounting and Economics* 57(1), 76–88.
- Konchitchki, Y. and P. N. Patatoukas (2014b). Taking the pulse of the real economy using financial statement analysis: Implications for macro forecasting and stock valuation. *Accounting Review* 89(2), 669–694.
- Lang, L., E. Ofek, and René M. Stulz (1996). Leverage, investment, and firm growth. *Journal of Financial Economics* 40(1), 3–29.
- Love, I. and L. Zicchino (2006). Financial development and dynamic investment behavior: Evidence from panel VAR. *Quarterly Review of Economics and Finance* 46(2), 190–210.
- Marcuss, R. (2004). Corporate profits in the GDP accounts. Technical Report 0040, United States Department of Commerce, Economics and Statistics Administration, Bureau of Economic Analysis.
- Myers, S. C. (1977). Determinants of corporate borrowing. *Journal of Financial Economics* 5(2), 147–175.
- Nalewaik, J. J. (2010). The Income- and Expenditure-Side Estimates of U.S. Output Growth. *Brookings Papers on Economic Activity* 2010(1), 71–106.
- Nallareddy, S. and M. Ogneva (2017). Predicting restatements in macroeconomic indicators using accounting information. *Accounting Review* 92(2), 151–182.
- Orcutt, G. H., H. W. Watts, and J. B. Edwards (1968). Data Aggregation and Information Loss. *American Economic Review* 58(4), 773–787.
- Ottonello, P. and T. Winberry (2020). Financial Heterogeneity and the Investment Channel

- of Monetary Policy. *Econometrica* 88(6), 2473–2502.
- Papanikolaou, D. (2011). Investment Shocks and Asset Prices. *Journal of Political Economy* 119(4), 639–685.
- Shen, Z. and D. Xiu (2025, January). Can Machines learn weak signals? NBER Working Paper 33421, National Bureau of Economic Research.
- Shivakumar, L. and O. Urcan (2017). Why does aggregate earnings growth reflect information about future inflation? *Accounting Review* 92(6), 247–276.
- Stock, J. H. and M. W. Watson (2002). Has the Business Cycle Changed and Why? *NBER Macroeconomics Annual* 17, 159–218.
- Wright, J. H. (2019). Some observations on forecasting and policy. *International Journal of Forecasting* 35(3), 1186–1192.